

# Limits – 5

Recurrences  $x_{n+1} = f(x_n)$ : fixed points and stability

## Homework

1 сентября 2026 года

**Theoretical minimum.** Fixed points and their stability, the criterion  $|f'(L)| < 1$ ; monotone iterations and 2-cycles; the contraction principle and the resulting estimate of the rate; the neutral case and polynomial asymptotics; Newton's method as a recurrence with  $f'(L) = 0$  and quadratic convergence.

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1. Prove the contraction principle on an interval and show that the pointwise condition  $|f'(x)| < 1$  is not sufficient: construct  $f: \mathbb{R} \rightarrow \mathbb{R}$  with  $|f'| < 1$  everywhere and no fixed point.
2. Study  $x_{n+1} = \lambda x_n(1 - x_n)$  on  $[0, 1]$  for  $0 < \lambda \leq 3$ : find the fixed points, determine their type and prove convergence; explain what happens at  $\lambda = 3$  and why a 2-cycle appears for  $\lambda > 3$ .
3. Let  $x_{n+1} = \frac{1}{2}(x_n + \frac{a}{x_n})$  with  $a > 0$  and  $x_0 > 0$ . Prove the quadratic convergence  $|x_{n+1} - \sqrt{a}| \leq C|x_n - \sqrt{a}|^2$  and explain it through  $f'(\sqrt{a}) = 0$ .
4. Let  $f$  be continuous on  $[0, 1]$  with  $f([0, 1]) \subseteq [0, 1]$ , and suppose the iterates converge for every initial value. Prove that the limit is independent of the starting point if and only if the fixed point is unique.
5. Let  $x_{n+1} = x_n + \frac{1}{x_n}$  with  $x_1 > 0$ . Prove that  $x_n \rightarrow \infty$  and find the asymptotics of  $x_n$  up to  $o(1)$ .
6. Let  $0 < x_1 < 1$  and  $x_{n+1} = x_n(1 - x_n)$ . Find  $\lim nx_n$  and the next term of the asymptotics  $x_n = \frac{1}{n} + \frac{c \ln n}{n^2} + o(\frac{\ln n}{n^2})$ .
7. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be continuous and suppose the sequence  $x_{n+1} = f(x_n)$  is bounded. Prove that its set of partial limits is connected and invariant under  $f$ .

8. Give an example of a continuous  $f: [0, 1] \rightarrow [0, 1]$  possessing a 3-cycle, explain what Sharkovsky's theorem has to do with it, and write down the Sharkovsky order for the first ten integers.
9. Let  $a_1 > 0$  and  $a_{n+1} = a_n + \frac{1}{a_n^2}$ . Prove that  $a_n \sim (3n)^{1/3}$  and refine the asymptotics to the second term.
10. Let  $f$  be differentiable with  $f(L) = L$  and  $f'(L) = -1$ . Prove that the iterates may either converge or diverge by giving both examples, and explain which criterion decides the issue (look at  $f \circ f$  and at its third derivative).